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NIMCET PYQ

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By Amit Katiyar (MCA-JNU)

1. If $a+b+c \neq 0$, then the system of equations
$(b+c)(y+z)-ax=b-c$
$(c+a)(z+x)-by=c-a$
$(a+b)(x+y)-cz=a-b$ has

NIMCET 2009

(a)a unique solution
(b)no solution
(c)infinite number of solutions
(d)finitely many solutions

2. If $a \neq p, b \neq q, c \neq r$ and $\begin{vmatrix} p & b & c \\ a & q & c \\ a & b & r \end{vmatrix} = 0$ then the value of $\frac{p}{p-a}+\frac{q}{q-b}+\frac{r}{r-c}$ is

NIMCET 2010

(a)0 (b)1 (c)-1 (d)2

3. Consider the system of linear equations
$3x_1+7x_2+x_3=2, x_1+2x_2+x_3=3$
$2x_1+3x_2+4x_3=13$
The system has

NIMCET 2011

(a)infinitely many solutions
(b)exactly 3 solutions
(c)a unique solutions
(d)no solutions

4. If ω is the cube root of unity, then the system of equations $x+\omega^2y+\omega z=0$ and $\omega x+y+\omega^2 z=0$ and $\omega^2 x+\omega y+\omega z=0$ is

NIMCET 2012

(a)consistent and has unique solutions
(b)consistent and has more than one solution
(c)inconsistent
(d)None of the above

5. The value of k for which the set of equations $3x+ky-2z=0$, $x+ky+3z=0$ and $2x+3y-4z=0$ has a non-trivial solution, is

NIMCET 2012

(a)$\frac{15}{2}$ (b)$\frac{17}{2}$ (c)$\frac{31}{2}$ (d)$\frac{33}{2}$

6. The number of values of k for which the system of equations $(k+1)x+8y=4k$ and $kx+(k+3)y=3k-1$ has infinitely many solutions, is

NIMCET 2012

(a)0 (b)1 (c)2 (d)infinite

7. If ω is cube root of unity, then find the value of
$$\begin{vmatrix} 1+\omega & \omega^2 & -\omega \\ 1+\omega^2 & \omega & -\omega^2 \\ \omega^2+\omega & \omega & -\omega^2 \end{vmatrix}$$

NIMCET 2013

(a)3ω (b)-3ω (c)$3\omega^2$ (d)$-3\omega^2$

8. If $A+B+C=\pi$, then the value of
$\begin{vmatrix} \sin(A+B+C) & \sin B & \cos C \\ -\sin B & 0 & \tan A \\ \cos(A+B) & -\tan A & 0 \end{vmatrix}$ is

NIMCET 2015

(a)0 (b)1
(c)$2\sin A\sin B$ (d)2

9. If a, b, c are the roots of the equation $x^3-3x^2+3x+7=0$, then the value of $\begin{vmatrix} 2bc-a^2 & c^2 & b^2 \\ c^2 & 2ac-b^2 & a^2 \\ b^2 & a^2 & 2ab-c^2 \end{vmatrix}$ is

NIMCET 2018

(a)9 (b)27 (c)81 (d)0

10. If x, y, z are distinct real numbers and
$\begin{vmatrix} x & x^2 & 2+x^3 \\ y & y^2 & 2+y^3 \\ z & z^2 & 2+z^3 \end{vmatrix} = 0$, then $xyz=$

NIMCET 2019

(a)1 (b)-1 (c)2 (d)-2

11. The number of values of k for which the linear equation $4x+ky+z=0$, $kx+4y+z=0$, $2x+2y+z=0$ possess a non-zero solution is

NIMCET 2020

(a)2 (b)1 (c)0 (d)3

12. If the system of equations $3x-y+4z=3$, $x+2y-3z=-2, 6x+5y+\lambda z=-3$ has atleast one solution then $\lambda=$?.

NIMCET 2021

(a)-5 (b)3 (c)5 (d)6

13. If $D=\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2+x & 1 \\ 1 & 1 & 2+y \end{vmatrix}$, $x \neq 0, y \neq 0$, then D is

NIMCET 2022

(a) divisible by x and y
(b) divisible by x but not by y
(c) divisible by $(x+1)$ and $(y+1)$
(d) divisible by $(1+x)$ but not $(1+y)$

14. **If** $A=\begin{vmatrix} 1 & 2 & 15 \\ 3 & 4 & 11 \\ 5 & 6 & 7 \end{vmatrix}$, let $\lambda_1, \lambda_2, \lambda_3$ be its eigen value then $(1+\lambda_1)(1+\lambda_2)(1+\lambda_3)=$ (Determinant)
(a)0 (b)-95 (c)-4 (d)12

NIMCET 2024

15. For an invertible matrix A, which of the following is not always true: (Determinant)
(a) |adj (A)| ≠0 (b) |A| ≠ 0
(c) $|AA^{-1}|$ = 1 (d) |A(adj(A))| ≠ 1

NIMCET 2024

16. if x, y and z are the three cube roots of 27, then the determinant of the matrix $\begin{bmatrix} x & y & z \\ y & z & x \\ z & x & y \end{bmatrix}$ is

(a) 0 (b) -1

(c) 1 (d) (x – y)(x – z)(y – z)

NIMCET 2025

ANSWER KEY

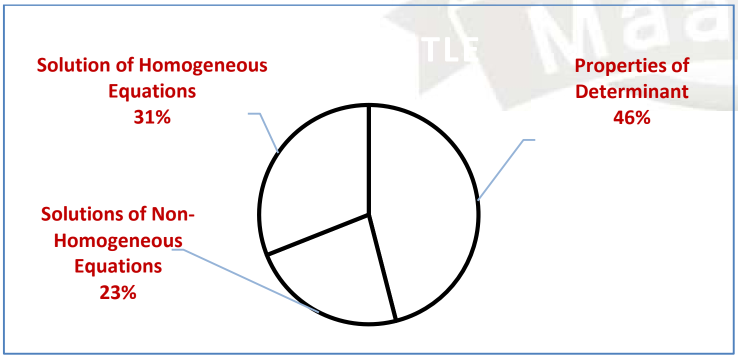
1.	a	2.	d	3.	a	4.	b	5.	d
6.	b	7.	d	8.	a	9.	d	10.	d
11.	a	12.	a	13.	c	14.	b	15.	d
16.	a								

Analysis 2007 to 2025

Questions Asked in NIMCET 2007 to 2025

Years	No. of Questions	Years	No. of Questions
NIMCET 2007	00	NIMCET 2016	00
NIMCET 2008	00	NIMCET 2017	00
NIMCET 2009	01	NIMCET 2018	01
NIMCET 2010	01	NIMCET 2019	01
NIMCET 2011	01	NIMCET 2020	01
NIMCET 2012	03	NIMCET 2021	01
NIMCET 2013	01	NIMCET 2022	01
NIMCET 2014	00	NIMCET 2023	00
NIMCET 2015	01	NIMCET 2024	02
		NIMCET 2025	01

Sub Topic Analysis



Sub Topicwise Analysis

Sub Topic	Question Asked
Properties of Determinant	46%
Solutions of Non-Homogeneous Equations	23%
Solution of Homogeneous Equations	31%

SOLUTION

1. (a)

$-ax+(b+c)y+(b+c)z=b-c$

$(c+a)x-by+(c+a)z=c-a$

$(a+b)x+(a+b)y-cz=a-b$

$$\begin{vmatrix} -a & b+c & b+c \\ c+a & -b & c+a \\ a+b & a+b & -c \end{vmatrix}$$

$R_1 \to R_1+R_2+R_3$

$$=(a+b+c)\begin{vmatrix} 1 & 1 & 1 \\ c+a & -b & c+a \\ a+b & a+b & -c \end{vmatrix}$$

$C_2 \to C_2-C_1$

$C_3 \to C_3-C_1$

$$=(a+b+c)\begin{vmatrix} 1 & 0 & 0 \\ c+a & -(a+b+c) & 0 \\ a+b & 0 & -(a+b+c) \end{vmatrix}$$

$=(a+b+c)^3 \neq 0$

So, system of equation has unique solution.

2. (d) $\begin{vmatrix} p & b & c \\ a & q & c \\ a & b & r \end{vmatrix}=0$

$R_2 \to R_2-R_1, R_3 \to R_3-R_1$

$$\begin{vmatrix} p & b & c \\ a-p & q-b & 0 \\ a-p & 0 & r-c \end{vmatrix}=0$$

$p(q-b)(r-c)-b(a-p)(r-c)-c(a-p)(q-b)=0$

$p(q-b)(r-c)+b(p-a)(r-c)+c(p-a)(q-b)=0$

Divided by $(p-a)(r-c)(q-b)$

$\frac{p}{p-a}+\frac{b}{q-b}+\frac{c}{r-c}=0$

$\Rightarrow \frac{p}{p-a}+\frac{b}{q-b}+1+\frac{c}{r-c}+1=2$

$\Rightarrow \frac{p}{p-a}+\frac{b+q-b}{q-b}+\frac{c+r-c}{r-c}=2$

$\Rightarrow \frac{p}{p-a}+\frac{q}{q-b}+\frac{r}{r-c}=2$

3. (a) $\Delta=\begin{vmatrix} 3 & 7 & 1 \\ 1 & 2 & 1 \\ 2 & 3 & 4 \end{vmatrix}=0$

Now we have to calculate $\Delta_{x_1}, \Delta_{x_2}, \Delta_{x_3}$

The value of $\Delta_{x_1}=\Delta_{x_2}=\Delta_{x_3}=0$

So, the system of equation has infinitely many solutions

4 (b) It is system of homogeneous equation.

$$\Delta=\begin{vmatrix} 1 & \omega^2 & \omega \\ \omega & 1 & \omega^2 \\ \omega^2 & \omega & 1 \end{vmatrix}$$

$C_1 \to C_1+C_2+C_3$

$$=\begin{vmatrix} 1+\omega+\omega^2 & \omega^2 & \omega \\ 1+\omega+\omega^2 & 1 & \omega^2 \\ 1+\omega+\omega^2 & \omega & 1 \end{vmatrix}=0\ [\because 1+\omega+\omega^2=0]$$

Here $\Delta=0$ it means system has non trivial solution (infinite solution)

So, the system of equation is consistent has more than one solution.

5 **(d)** It is system of homogeneous equation for non trivial solution (infinite solution) $\Delta = 0$.

$$\begin{vmatrix} 3 & k & -2 \\ 1 & k & 3 \\ 2 & 3 & -4 \end{vmatrix} = 0$$

$= 3(-4k-9) - k(-4-6) - 2(3-2k) = 0$

$\Rightarrow$ k = $= \frac{33}{2}$

6. **(b)** There are two straight lines for infinite solutions these straight lines must be coincident

$a_1x + b_1y + c_1 = 0$

$a_2x + b_2y + c_2 = 0$

$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

$\Rightarrow \frac{k+1}{k} = \frac{8}{k+3} = \frac{4k}{3k-1}$

Solving these three equations we get $k = 1{,}3$ But only $k = 1$ satisfy all the equations.

7 **(d).** Since, ω is cube root of unity. $1 + \omega + \omega^2 = 0$ and $\omega^3 = 1$

Now, $\begin{vmatrix} 1+\omega & \omega^2 & -\omega \\ 1+\omega^2 & \omega & -\omega^2 \\ \omega^2+\omega & \omega & -\omega^2 \end{vmatrix}$

$C_1 \to C_1 + C_2$

$$= \begin{vmatrix} 0 & \omega^2 & -\omega \\ 0 & \omega & -\omega^2 \\ \omega^2+2\omega & \omega & -\omega^2 \end{vmatrix}$$

$R_3 \to R_3 - R_2$

$$= \begin{vmatrix} 0 & \omega^2 & -\omega \\ 0 & \omega & -\omega^2 \\ \omega^2+2\omega & 0 & 0 \end{vmatrix}$$

$= (\omega^2 + 2\omega)(-\omega^4 + \omega^2) = (\omega^2 + \omega + \omega)(-\omega + \omega^2)$

$= (-1 + \omega)(-\omega + \omega^2) = \omega - \omega^2 - \omega^2 + \omega^3$

$= 1 + \omega - 2\omega^2 = -\omega^2 - 2\omega^2 = -3\omega^2$

8 **(a).** $\begin{vmatrix} \sin\pi & \sin B & \cos C \\ -\sin B & 0 & \tan A \\ \cos(\pi - C) & -\tan A & 0 \end{vmatrix}$

$$= \begin{vmatrix} 0 & \sin B & \cos C \\ -\sin B & 0 & \tan A \\ -\cos C & -\tan A & 0 \end{vmatrix}$$

Now, this determinant is skew symmetric of odd order Hence, its value will be zero.

9. **(d)** Using Vieta's theorem for

$x^3 - 3x^2 + 3x + 7 = 0$ has roots a, b, c

$a + b + c = 3, ab + bc + ca = 3, abc = -7$

$$\begin{vmatrix} 2bc - a^2 & c^2 & b^2 \\ c^2 & 2ac - b^2 & a^2 \\ b^2 & a^2 & 2ab - c^2 \end{vmatrix}$$

$$= \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \begin{vmatrix} -a & c & b \\ -b & a & c \\ -c & b & a \end{vmatrix} = -\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \begin{vmatrix} a & c & b \\ b & a & c \\ c & b & a \end{vmatrix}$$

$$= \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

$= (3abc - a^3 - b^3 - c^3)^2$

We know that,

$(a + b + c)^3 = a^3 + b^3 + c^3 + 3(a + b + c)$

$(ab + bc + ca) - 3abc$

$\Rightarrow 27 = a^3 + b^3 + c^3 + 3(3)(3) + 3 \times 7$

$\Rightarrow a^3 + b^3 + c^3 = -21$

So, $(3abc - a^3 - b^3 - c^3)^2 = [3(-7) + 21]^2 = 0$

10 **(d).** $\begin{vmatrix} x & x^2 & 2 \\ y & y^2 & 2 \\ z & z^2 & 2 \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ y & y^2 & y^3 \\ z & z^2 & z^3 \end{vmatrix} = 0$

$$\Rightarrow 2\begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} + xyz\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} + xyz\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = 0$$

$$\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} [2 + xyz] = 0$$

$\Rightarrow xyz = -2$

11. **(a)** It is system of homogeneous equations.

So, for non-zero solution, $\Delta = 0$.

$$\Rightarrow \begin{vmatrix} 4 & k & 1 \\ k & 4 & 1 \\ 2 & 2 & 1 \end{vmatrix} = 0$$

$\Rightarrow 4(4-2) - k(k-2) + 1(2k-8) = 0$

$\Rightarrow 8 - k^2 + 2k + 1(2k-8) = 0$

$\Rightarrow k^2 - 4k = 0$

$\Rightarrow k(k-4) = 0 \Rightarrow k = 0{,}4$(2 values)

12 **(a).** $\Delta = 0$

$$\begin{vmatrix} 3 & -1 & 4 \\ 1 & 2 & -3 \\ 6 & 5 & \lambda \end{vmatrix} = 0$$

$3(2\lambda + 15) + 1(\lambda + 18) + 4(5 - 12) = 0$

$\Rightarrow 6\lambda + 45 + \lambda + 18 + 20 - 48 = 0$

$\Rightarrow 7\lambda = -35 \Rightarrow \lambda = -5$

13. (c) $D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2+x & 1 \\ 1 & 1 & 2+y \end{vmatrix}$

$= [(x+2)(y+2)-1]-[y+2-1]+1[1-x-2]$

$= (xy+2x+2y+4-1)-y-1+1-x-2$

$= xy+x+y+1$

$= (1+x)(1+y)$

14. (b)

. $A = \begin{bmatrix} 1 & 2 & 15 \\ 3 & 4 & 11 \\ 5 & 6 & 7 \end{bmatrix}$, let $\lambda_1 \lambda_2, \lambda_3$ be eigen value

For eigen value

$|A - \lambda I| = 0$

$\begin{vmatrix} 1-\lambda & 2 & 15 \\ 3 & 4-\lambda & 11 \\ 5 & 6 & 7-\lambda \end{vmatrix} = 0$

$(1-\lambda)[(4-\lambda)(7-\lambda)-66]-2(21-3\lambda-55)+15(8-20+5\lambda)$

$(1-\lambda)(4-\lambda)(7-\lambda)-66+66\lambda+2\times 34+6\lambda+75\lambda+30$

$(1-\lambda)(4-\lambda)(7-\lambda)-66+68+130+72\lambda+75\lambda$

$28-\lambda^3+12\lambda^2-39\lambda+28+147\lambda=0$

$\lambda^3-12\lambda^2-39\lambda-108\lambda=0$

-95

15. (d)

A. $|\text{adj}(A)| \neq 0$

if a A is invertible matrix then $|A| \neq 0$

$|\text{adj } A| = |A|^{n-1}$

$|\text{adj } A| \neq 0$

$|A| \neq 0$

$|AA^{-1}| = I$

$|A(\text{adj}(A)| \neq L$

if $A = I$

then $|A| = 1$

$|A(\text{adj}(A)| \neq L$

so option d is incorrect.

16. (a)

the cube roots are $x = 3, \mathrm{y} = 3\mathrm{w}, \mathrm{z} = 3w^2$ with

$1 + \mathrm{w} + w^2 = 0,$

so $x + y + z = 0$. Hence the rows of the matrix are linearly dependent (their sum is the zero vector), so the determinant is 0.